ISLAMIC UNIVERSITY OF TECHNOLOGY

COURSE NO: EEE 4416

Project

on

TIC-TAC-TOE Game

Submitted by: ***Team Noob (Group-10)***

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Abstract:

This project involves the development of a Tic Tac Toe game application using MATLAB. The primary objective is to create an interactive and user-friendly game interface that allows two players to compete against each other in the classic game of Tic Tac Toe. Additionally, the project aims to enhance our understanding and proficiency in MATLAB programming, graphical user interface (GUI) design, and the implementation of game logic algorithms.

Objectives:

1. Develop a functional Tic Tac Toe game application in MATLAB.
2. Design an intuitive graphical user interface (GUI) for seamless user interaction.
3. Implement robust game logic to handle all possible game scenarios, including win, draw, and invalid moves.
4. Ensure the application runs efficiently and provides a smooth user experience.

Methods:

1. **GUI Design:** Using MATLAB's GUI development environment (GUIDE) to create a visually appealing and easy-to-use interface. This includes designing buttons for each cell of the Tic Tac Toe grid and indicators for player turns.
2. **Game Logic Implementation:** Writing MATLAB scripts to handle game rules, determine valid moves, check for win conditions, and manage game states (e.g., ongoing game, player win, draw).
3. **Event Handling:** Implementing callback functions that respond to user inputs, such as clicking on a grid cell, to update the game state and interface dynamically.
4. **Testing and Debugging:** Conducting extensive testing to identify and fix bugs, ensuring the application handles edge cases and unexpected user inputs gracefully.

Key Findings:

1. The final MATLAB application successfully allows two players to play Tic Tac Toe with clear visual indicators of the game state.
2. The GUI is intuitive and provides real-time feedback to the players, enhancing the overall user experience.
3. The game logic effectively manages all possible game scenarios, ensuring accurate and fair gameplay.
4. The project improved our skills in MATLAB programming, particularly in GUI design and event-driven programming.

This project demonstrates the potential of MATLAB as a tool for developing simple interactive applications and highlights the importance of thorough testing and user-centered design in software development.

Introduction:

The motivation behind our project is to learn and practice programming by creating a Tic Tac Toe game in MATLAB. Tic Tac Toe is a simple and well-known game, making it a good choice for a beginner's project. Our main goals are:

1. **Learning MATLAB:** Understanding how to use MATLAB for programming and creating graphical user interfaces (GUIs).
2. **Practical Experience:** Gaining hands-on experience in writing code, finding and fixing errors, and making the game work correctly.
3. **Game Logic:** Implementing the rules of Tic Tac Toe, such as determining wins, draws, and valid moves.
4. **User Interface:** Designing a simple and easy-to-use interface for players.

Tic Tac Toe Game Rules:

Tic Tac Toe is a simple game played on a 3x3 grid. Here are the basic rules:

a) **Players:** The game is for two players. One uses "X" and the other uses "O."

b) **Objective:** The goal is to get three of your marks (either "X" or "O") in a row. The row can be horizontal, vertical, or diagonal.

c) **Gameplay:**
   - Players take turns placing their mark in an empty cell on the 3x3 grid.
   - You can place your mark in any empty cell.

d) **Winning the Game:**

One will win by getting three of his marks in a row (horizontal, vertical, or diagonal).

The game ends as soon as one player wins.

e) Draw:

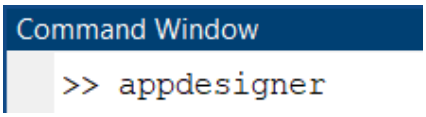
If all nine cells are filled and no one has three in a row, the game is a draw.

f) Invalid Moves:

A move is invalid if one try to place his mark in a cell that is already taken.

Methodology & Algorithm:

In this project we have use Matlab App Buider to make a GUI or Graphical User Interface. Firstly we gave command to matlab command window 'appdesigner'. The matlab app builder page opened. Then we took a blank app to get a blank space for making our project.



We then took 9 push Button from the component library. All the buttons are resized as square. Then all the buttons are arranged in 3 column and 3 rows to make the interface of a tick-tack-toe game.

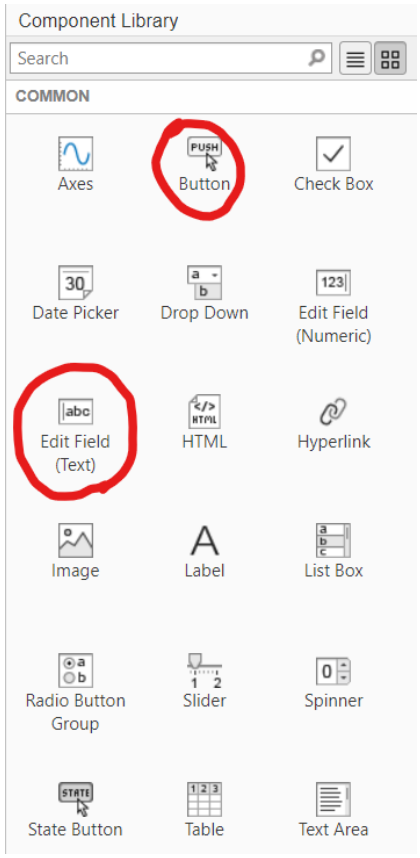


Fig. Component Library

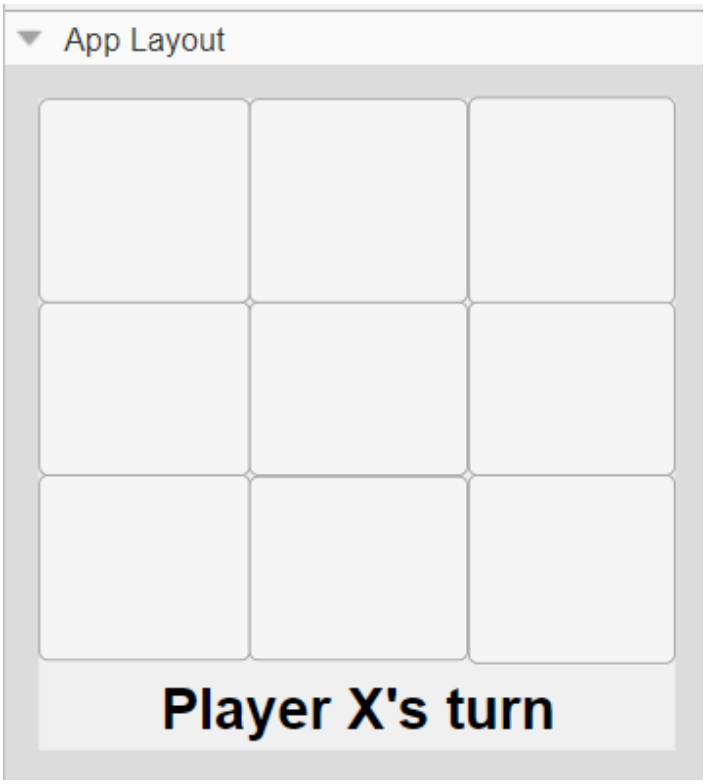


Fig. App Layout

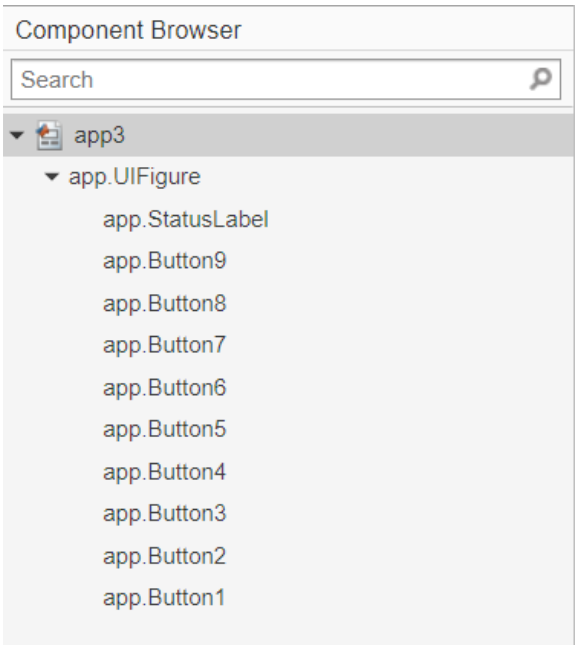


Fig. Component Browser

So, the board is defined as- board = zeros(3, 3);
Here, by arranging the buttons by default each buttons have got a row and column number according to their position. The board properties is set to 0. It means the board is empty, no player pushed any button. currentPlayer is set to 1 means current player is X. 2 is for player O.

Then a initialize function is added, where status of the board is updated by updateStatus(). This function shows a text of which players turn is at present. For showing the text we have added a text field below 3*3 buttons table and named it StatusLabel It currentPlayer variable is 1 then it's X's turn else if the variable's value is 2 then it's player Y's turn.

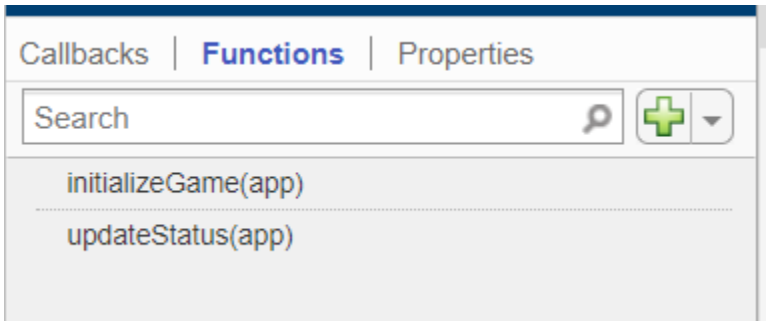


Fig. Functions

So, on the initialize function each of the Buttons text is set to blank and all the buttons are enabled.

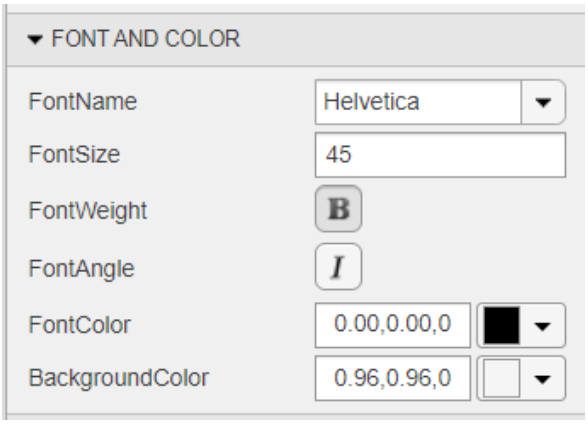


Fig. Font & Color of a Button.

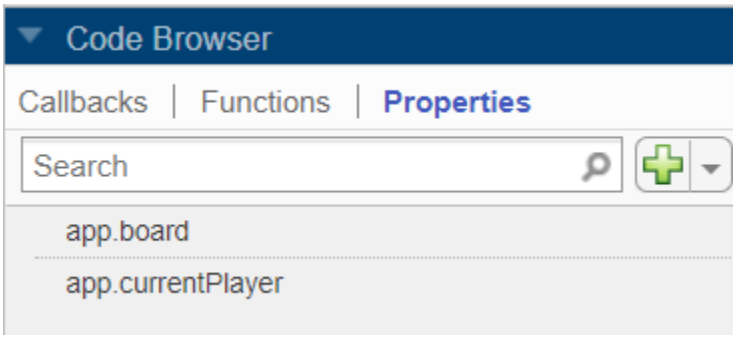


Fig. Properties

Then for each button a call back function is added. When one of the button is pushed this callback function will check if there is any existing text one this button. If there is not then it will give a text based on the currentPlayer value. Then the status will be updated bu updateStatus() function. Then this function will check any winning combination is found or not.

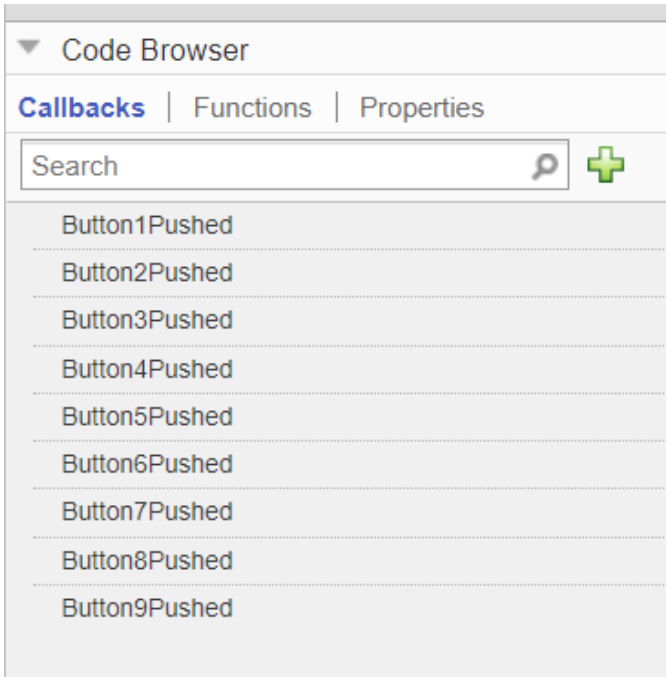


Fig. Callbacks

On the checkWinner function there is 8 winning combination is set which is-

```
winningCombinations = [1 2 3; 4 5 6; 7 8 9; 1 4 7; 2 5 8; 3 6 9; 1 5 9; 3 5 7];
```

Then a for loop is used to go through every row of this winning combination matrix. There is two condition inside this loop. In the first condition, this function will check is any of the winning combination in board is 1, is so then it will declare in the statusLabel that ‘Player X wins!’ and then it will disable all the buttons of the board. Other condition is if any of the winning combination in board is 2, is so then it will declare in the statusLabel that ‘Player Y wins!’ and then it will disable all the buttons of the board. If all the buttons is not 0 and no winning combination is found then it will update status level as ‘It’s a draw!’

Disable of buttons will be held by disableButtons() function. In this function it will loop thorugh all the Button and make Enable value ‘off’

So, this is a brief explanation of the code and functions used in the code to make a simple app with graphical use interface using Matlab App Builder.

Code:

```
classdef app3 < matlab.apps.AppBase
```

```
    % Properties that correspond to app components
    properties (Access = public)
        UIFigure     matlab.ui.Figure
        StatusLabel  matlab.ui.control.Label
        Button9      matlab.ui.control.Button
        Button8      matlab.ui.control.Button
        Button7      matlab.ui.control.Button
        Button6      matlab.ui.control.Button
        Button5      matlab.ui.control.Button
        Button4      matlab.ui.control.Button
        Button3      matlab.ui.control.Button
        Button2      matlab.ui.control.Button
        Button1      matlab.ui.control.Button
    end
```

```
    properties (Access = private)
        board = zeros(3, 3); % Game board (0 = empty, 1 = X, 2 = O)
        currentPlayer = 1; % Current player (1 = X, 2 = O) % Description
    end
```

```
    methods (Access = private)
```

```
        function initializeGame(app)
            app.board = zeros(3, 3);
            app.currentPlayer = 1;
            app.updateStatus();
            for i = 1:9
                app.(['Button', num2str(i)]).Text = '';
                app.(['Button', num2str(i)]).Enable = 'on';
            end
        end
```

```
        function updateStatus(app)
            if app.currentPlayer == 1
                app.StatusLabel.Text = 'Player X''s turn';
            else
                app.StatusLabel.Text = 'Player O''s turn';
            end
        end
    end
```

```
    % Callbacks that handle component events
    methods (Access = private)
```

```
        % Button pushed function: Button1
        function Button1Pushed(app, event)
            buttonNumber = 1; % Assuming this button is associated with button
number 1
            [row, col] = ind2sub([3, 3], buttonNumber);
```

```
            if app.board(row, col) == 0
                app.board(row, col) = app.currentPlayer;
                if app.currentPlayer == 1
                    app.(['Button', num2str(buttonNumber)]).Text = 'X';
                    app.currentPlayer = 2;
                else
                    app.(['Button', num2str(buttonNumber)]).Text = 'O';
                    app.currentPlayer = 1;
                end
                app.updateStatus();
                app.checkWinner();
            end
```

```
        end
```

```
        % Button pushed function: Button2
        function Button2Pushed(app, event)
            buttonNumber=2;
            [row, col] = ind2sub([3, 3], buttonNumber);
```

```
            if app.board(row, col) == 0
                app.board(row, col) = app.currentPlayer;
                if app.currentPlayer == 1
                    app.(['Button', num2str(buttonNumber)]).Text = 'X';
                    app.currentPlayer = 2;
                else
                    app.(['Button', num2str(buttonNumber)]).Text = 'O';
                    app.currentPlayer = 1;
                end
                app.updateStatus();
```

```
                app.checkWinner();

            end

        end

        % Button pushed function: Button3
        function Button3Pushed(app, event)
            buttonNumber=3;
            [row, col] = ind2sub([3, 3], buttonNumber);

            if app.board(row, col) == 0
                app.board(row, col) = app.currentPlayer;
                if app.currentPlayer == 1
                    app.(['Button', num2str(buttonNumber)]).Text = 'X';
                    app.currentPlayer = 2;
                else
                    app.(['Button', num2str(buttonNumber)]).Text = 'O';
                    app.currentPlayer = 1;
                end
                app.updateStatus();

                app.checkWinner();

            end

        end

        % Button pushed function: Button4
        function Button4Pushed(app, event)
            buttonNumber=4;
            [row, col] = ind2sub([3, 3], buttonNumber);

            if app.board(row, col) == 0
                app.board(row, col) = app.currentPlayer;
                if app.currentPlayer == 1
                    app.(['Button', num2str(buttonNumber)]).Text = 'X';
                    app.currentPlayer = 2;
                else
```

```
                app.(['Button', num2str(buttonNumber)]).Text = 'O';
                app.currentPlayer = 1;
            end
            app.updateStatus();


            app.checkWinner();


        end


    end


    % Button pushed function: Button5
    function Button5Pushed(app, event)
        buttonNumber=5;
        [row, col] = ind2sub([3, 3], buttonNumber);


        if app.board(row, col) == 0
            app.board(row, col) = app.currentPlayer;
            if app.currentPlayer == 1
                app.(['Button', num2str(buttonNumber)]).Text = 'X';
                app.currentPlayer = 2;
            else
                app.(['Button', num2str(buttonNumber)]).Text = 'O';
                app.currentPlayer = 1;
            end
            app.updateStatus();


            app.checkWinner();
        end




    end


    % Button pushed function: Button6
    function Button6Pushed(app, event)
        buttonNumber=6;
        [row, col] = ind2sub([3, 3], buttonNumber);


        if app.board(row, col) == 0
            app.board(row, col) = app.currentPlayer;
```

```
            if app.currentPlayer == 1
                app.(['Button', num2str(buttonNumber)]).Text = 'X';
                app.currentPlayer = 2;
            else
                app.(['Button', num2str(buttonNumber)]).Text = 'O';
                app.currentPlayer = 1;
            end
            app.updateStatus();


            app.checkWinner();
        end


    end


    % Button pushed function: Button7
    function Button7Pushed(app, event)
        buttonNumber=7;
        [row, col] = ind2sub([3, 3], buttonNumber);


        if app.board(row, col) == 0
            app.board(row, col) = app.currentPlayer;
            if app.currentPlayer == 1
                app.(['Button', num2str(buttonNumber)]).Text = 'X';
                app.currentPlayer = 2;
            else
                app.(['Button', num2str(buttonNumber)]).Text = 'O';
                app.currentPlayer = 1;
            end
            app.updateStatus();


            app.checkWinner();


        end


    end


    % Button pushed function: Button8
    function Button8Pushed(app, event)
        buttonNumber=8;
        [row, col] = ind2sub([3, 3], buttonNumber);
```

```
            if app.board(row, col) == 0
                app.board(row, col) = app.currentPlayer;
                if app.currentPlayer == 1
                    app.(['Button', num2str(buttonNumber)]).Text = 'X';
                    app.currentPlayer = 2;
                else
                    app.(['Button', num2str(buttonNumber)]).Text = 'O';
                    app.currentPlayer = 1;
                end
                app.updateStatus();

                app.checkWinner();

            end

        end

        % Button pushed function: Button9
        function Button9Pushed(app, event)
            buttonNumber=9;
            [row, col] = ind2sub([3, 3], buttonNumber);

            if app.board(row, col) == 0
                app.board(row, col) = app.currentPlayer;
                if app.currentPlayer == 1
                    app.(['Button', num2str(buttonNumber)]).Text = 'X';
                    app.currentPlayer = 2;
                else
                    app.(['Button', num2str(buttonNumber)]).Text = 'O';
                    app.currentPlayer = 1;
                end
                app.updateStatus();
                app.checkWinner();

            end
        end

        % Check for winner
        function checkWinner(app)
            winningCombinations = [1 2 3; 4 5 6; 7 8 9; 1 4 7; 2 5 8; 3 6 9; 1
5 9; 3 5 7];
            for i = 1:size(winningCombinations, 1)
                if all(app.board(winningCombinations(i, :)) == 1)
```

```
                    app.StatusLabel.Text = 'Player X wins!';
                    app.disableButtons();
                    return;
                elseif all(app.board(winningCombinations(i, :)) == 2)
                    app.StatusLabel.Text = 'Player O wins!';
                    app.disableButtons();
                    return;
                end
            end

            if all(app.board(:) ~= 0)
                app.StatusLabel.Text = 'It''s a draw!';
            end
        end

        % Disable buttons
        function disableButtons(app)
            for i = 1:9
                app.(['Button', num2str(i)]).Enable = 'off';
            end

        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create UIFigure and hide until all components are created
            app.UIFigure = uifigure('Visible', 'off');
            app.UIFigure.Position = [100 100 337 344];
            app.UIFigure.Name = 'MATLAB App';

            % Create Button1
            app.Button1 = uibutton(app.UIFigure, 'push');
            app.Button1.ButtonPushedFcn = createCallbackFcn(app,
@Button1Pushed, true);
            app.Button1.FontSize = 45;
            app.Button1.FontWeight = 'bold';
```

```
            app.Button1.Position = [1 237 112 108];
            app.Button1.Text = '';
```

```
            % Create Button2
            app.Button2 = uibutton(app.UIFigure, 'push');
            app.Button2.ButtonPushedFcn = createCallbackFcn(app,
@Button2Pushed, true);
            app.Button2.FontSize = 45;
            app.Button2.FontWeight = 'bold';
            app.Button2.Position = [112 237 116 108];
            app.Button2.Text = '';
```

```
            % Create Button3
            app.Button3 = uibutton(app.UIFigure, 'push');
            app.Button3.ButtonPushedFcn = createCallbackFcn(app,
@Button3Pushed, true);
            app.Button3.FontSize = 45;
            app.Button3.FontWeight = 'bold';
            app.Button3.Position = [228 236 110 110];
            app.Button3.Text = '';
```

```
            % Create Button4
            app.Button4 = uibutton(app.UIFigure, 'push');
            app.Button4.ButtonPushedFcn = createCallbackFcn(app,
@Button4Pushed, true);
            app.Button4.FontSize = 45;
            app.Button4.FontWeight = 'bold';
            app.Button4.Position = [1 146 112 92];
            app.Button4.Text = '';
```

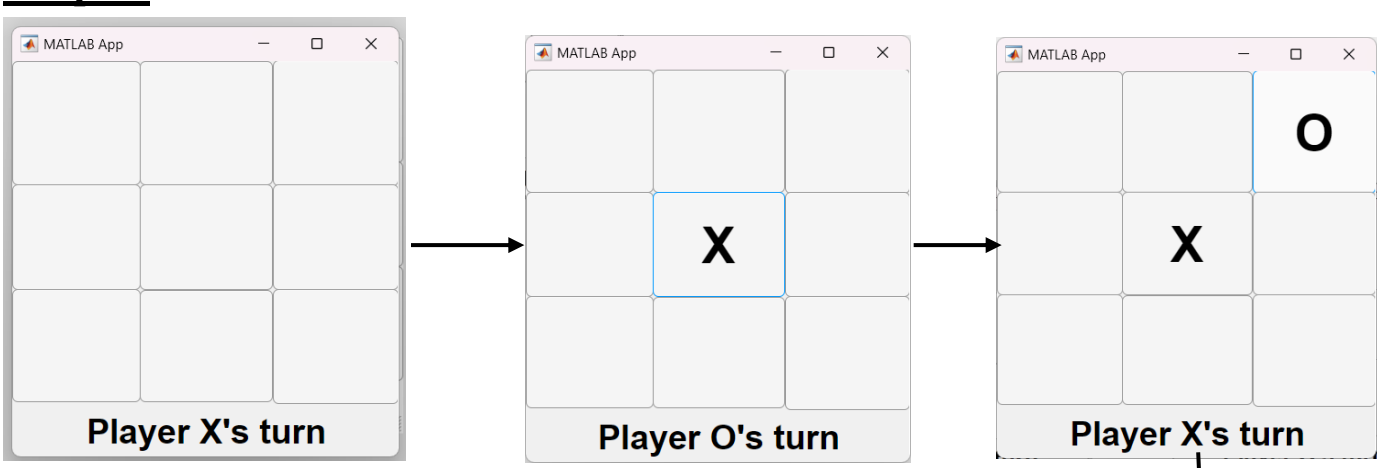
```
            % Create Button5
            app.Button5 = uibutton(app.UIFigure, 'push');
            app.Button5.ButtonPushedFcn = createCallbackFcn(app,
@Button5Pushed, true);
            app.Button5.FontSize = 45;
            app.Button5.FontWeight = 'bold';
            app.Button5.Position = [112 146 116 92];
            app.Button5.Text = '';
```

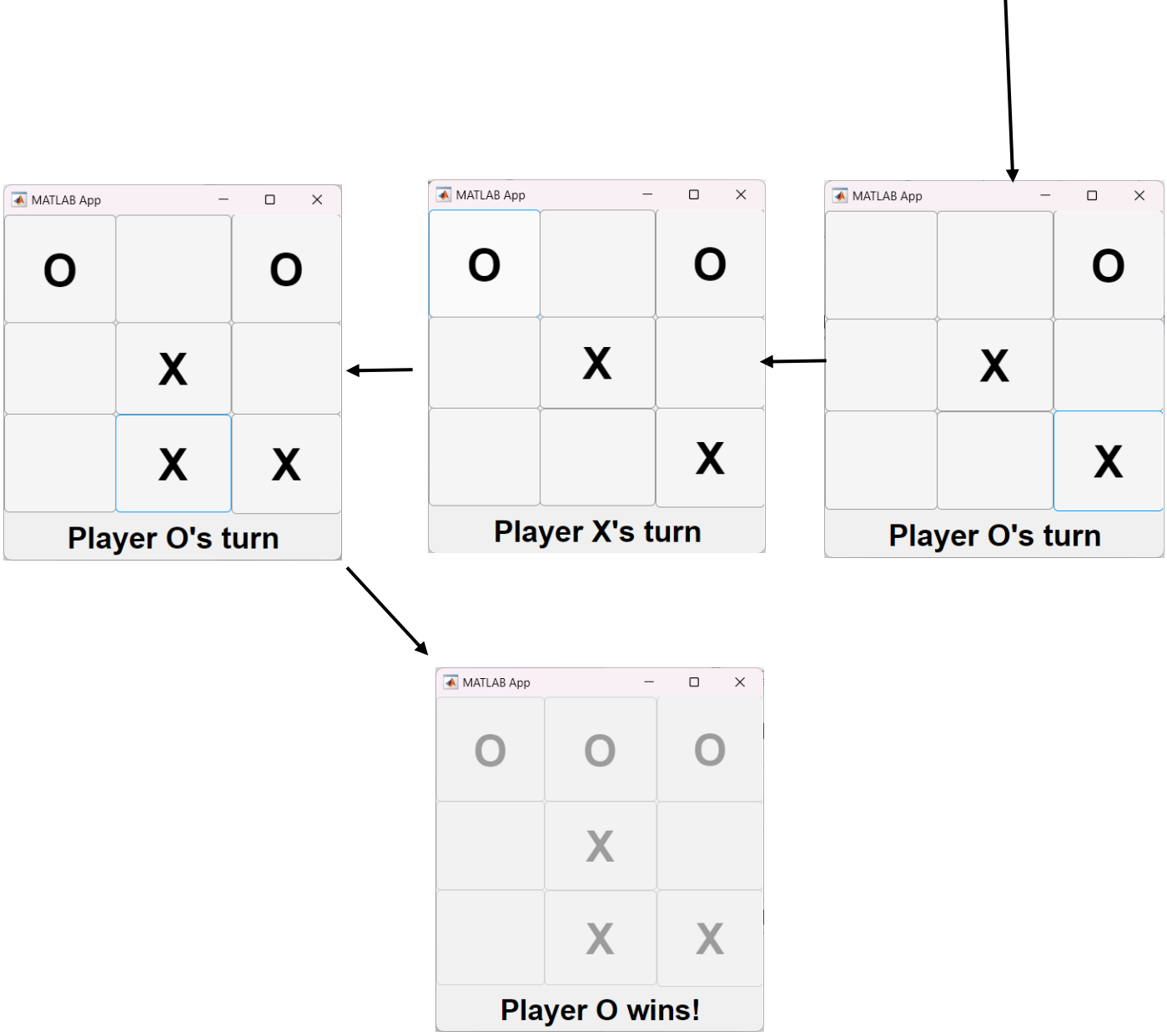
```
            % Create Button6
            app.Button6 = uibutton(app.UIFigure, 'push');
            app.Button6.ButtonPushedFcn = createCallbackFcn(app,
@Button6Pushed, true);
            app.Button6.FontSize = 45;
            app.Button6.FontWeight = 'bold';
```

```
            app.Button6.Position = [228 146 110 92];
            app.Button6.Text = '';

            % Create Button7
            app.Button7 = uibutton(app.UIFigure, 'push');
            app.Button7.ButtonPushedFcn = createCallbackFcn(app,
@Button7Pushed, true);
            app.Button7.FontSize = 45;
            app.Button7.FontWeight = 'bold';
            app.Button7.Position = [1 48 112 99];
            app.Button7.Text = '';

            % Create Button8
            app.Button8 = uibutton(app.UIFigure, 'push');
            app.Button8.ButtonPushedFcn = createCallbackFcn(app,
@Button8Pushed, true);
            app.Button8.FontSize = 45;
            app.Button8.FontWeight = 'bold';
            app.Button8.Position = [112 48 116 98];
            app.Button8.Text = '';

            % Create Button9
            app.Button9 = uibutton(app.UIFigure, 'push');
            app.Button9.ButtonPushedFcn = createCallbackFcn(app,
@Button9Pushed, true);
            app.Button9.FontSize = 45;
            app.Button9.FontWeight = 'bold';
            app.Button9.Position = [228 47 110 100];
            app.Button9.Text = '';

            % Create StatusLabel
            app.StatusLabel = uilabel(app.UIFigure);
            app.StatusLabel.HorizontalAlignment = 'center';
            app.StatusLabel.FontSize = 30;
            app.StatusLabel.FontWeight = 'bold';
            app.StatusLabel.Position = [62 -3 216 51];
            app.StatusLabel.Text = 'Player X''s turn';

            % Show the figure after all components are created
            app.UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
```

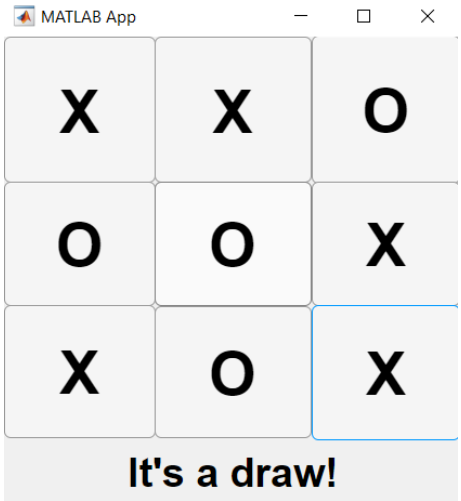
```
    methods (Access = public)

        % Construct app
        function app = app3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.UIFigure)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.UIFigure)
        end
    end
end
```

Output:





When a draw happens it shows:



Contributions:

Student ID	Percentage of contribution	Specific contribution
210021305	16%	Report Writing
210021337	30%	Coding+ Code Explanation
210021132	15%	Report Writing
210021134	20%	Report Writing +Code explanation
210021136	14%	Report Writing
210021149	5%	Cover Page